

MTH203: LECTURE 02

In this lecture, we will have briefly review some basic concepts and results from set theory. Our treatment of set theory is not completely rigorous, but will be adequate for our purposes.

1. SETS: BASIC NOTATION AND TERMINOLOGY

A *set* is a collection of objects. The objects in the set are called its *elements*. The statement that an element x is an element of a set S is symbolically written as $x \in S$. If x is not an element of S , this is written as $x \notin S$.

Two sets A and B are said to be *equal* (written as $A = B$) if and only if every element of A is also in B and every element of B is also in A .

Sets may be described by listing the elements. For example:

$$S = \{2, 4, 6, 8\}$$

Sometimes, a set is described in terms of a property that characterizes its elements. For example:

$$S = \{x | x \text{ is an integer and } 5 \leq x \leq 20\}$$

Sometimes, we use the elements of one set as labels for the elements of another set. For example, suppose a_1, a_2, a_3 are elements of a set S . Here, we are using 1, 2 and 3 essentially as labels. Let I denote the set $\{1, 2, 3\}$. Then, we may write S as $\{a_i\}_{i \in I}$. This is essentially another way of writing $\{a_i | i \in I\}$.

The *empty set* $\{\}$, which has no elements at all, will be denoted by $\emptyset$.

2. SUBSETS

Let A and B be two sets. We say that A is a *subset* of B if for every $x \in A$, we also have $x \in B$. This is expressed by the symbols $A \subset B$ or $A \subseteq B$. The statement that A is *not* a subset of B is written as $A \not\subseteq B$.

Observe that the empty set $\emptyset$ is a subset of any other set. Also, for any set S , we have $S \subset S$.

We say that A is a *proper subset* of B if $A \subset B$ but $A \neq B$. This is written symbolically as $A \subsetneq B$.

Example 1. Let $S = \{1, 2, 3\}$. Then, the subsets of S are as follows:

$$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}$$

The set of all subsets of a set S is called the *power set* of S and is denoted by $\mathcal{P}(S)$. In the above example, we saw that the set $\{1, 2, 3\}$, which has 3 elements, has $2^3 = 8$ subsets. In general, one can show that if a set has n elements, where n is any non-negative integer, then its power set has 2^n elements. (Do you see why?)

3. SOME CONSTRUCTIONS INVOLVING SETS

3.1. Union. Let A and B be two sets. The *union* $A \cup B$ is defined as

$$A \cup B := \{x | x \in A \text{ or } x \in B\}.$$

More generally, we can define the union of a family of sets. For example, suppose $\{A_s\}_{s \in S}$ is a collection of sets, indexed (i.e. labelled) by the set S . Then, we define the union of this family as

$$\bigcup_{s \in S} A_s := \{x | x \in A_s \text{ for some } s \in S\}.$$

Thus, in plain English, this is just the collection of all the elements that are in some A_s as s varies through all the elements of the set S . The set S can also be infinite in this case.

3.2. Intersection: Let A and B be two sets. The *intersection* $A \cap B$ is defined as

$$A \cap B := \{x | x \in A \text{ or } x \in B\}.$$

More generally, we can define the intersection of a family of sets. If $\{A_s\}_{s \in S}$ is a collection of sets, we define the intersection of this family as

$$\bigcap_{s \in S} A_s := \{x | x \in A_s \text{ for all } s \in S\}.$$

Thus, the intersection is the collection of all the elements that are in *all* the sets A_s as s varies through all the elements of the set S .

3.3. Cartesian products: Let A and B be two sets. The *cartesian* $A \times B$ is defined as the collection of all *ordered* pairs (a, b) where $a \in A$ and $b \in B$.

An ordered pair (a, b) consists of the two elements, along with the choice of a as the *first* element in the pair and b as the *second* element. It is clear that one can easily generalize this to products of the form $A_1 \times A_2 \times \cdots \times A_n$. A typical element of $A_1 \times \cdots \times A_n$ is an *n-tuple* of the form $(a_1, \dots, a_n)$ where $a_i \in A_i$ for all i .

More generally, we can also talk about the cartesian product of a family of sets $\{A_s\}_{s \in S}$. A typical element of this set is written as $(a_s)_{s \in S}$ where $a_s \in A_s$ for all $s \in S$. (Note that there does not need to be an order on the indexing set. The *tuple* $(a_s)_{s \in S}$ consists of not just the elements a_s , but also a record of the fact that a_s came from the set A_s .) Thus, the cartesian product can be seen as just a choice of *representatives* for each set A_s . Thus, the cartesian product $\prod_{s \in S} A_s$ can be seen as the set of all possible choices of representatives for the sets A_s . If S is infinite, it is not immediately clear that even a single choice actually exists. (After all, you do you specify infinitely many choices in general?) In fact, the statement that at least one such choice exists is called the *Axiom of Choice*. This is one of the basic axioms of set theory. (We will not really need such general cartesian products in this course. If you are curious, you may read about this in [1]).

4. RELATIONS AND FUNCTIONS

Let A and B be two sets. A *relation* R from A to B is a subset of the cartesian product $A \times B$. This is meant to encode the intuitive idea we have of a relationship between objects. The statement that the pair (a, b) lies in R can be written as aRb .

Example 2. Consider the statement, “Shekhar likes the colour red.” This is a relationship between the person Shekhar and the colour red. Suppose we have a set P of people and the set C of colours. Then we could collect all the information about which colours each person in the set P likes. How might we express such data concisely? The fact that Shekhar likes the colour red could just be expressed as the ordered pair (Shekhar, red). This could be done in general – we could record the information in terms of ordered pairs, the first entry being the person and the second being a colour that he or she likes. Thus, this relationship between people and colours can be represented as a subset of $P \times C$.

Sometimes, we want to talk about a relation R from a set A into itself. Then, we say that R is a relation *on* the set A .

Example 3. We denote the set of real numbers by $\mathbb{R}$. We will define a relation called $>$ on $\mathbb{R}$. A pair (x, y) of real numbers lies in $>$ if there exists a real number z such that $x - y = z^2$.

Clearly, there exists a real number z such that $x - y = z^2$ if and only if $x - y$ is *positive*. In other words, $>$ is the usual order relation on $\mathbb{R}$.

A function f from a set A into a set B is a relation from A to B such that for every element a of A , there is exactly one element b of B such that $(a, b) \in f$. Usually, this is written as $f(a) = b$ (instead of using the notation afb as we did for relations in general). The set A is called the *domain* of f and B is called the *codomain* of f .

Intuitively, a function is a “rule” which assigns a unique element of B to every element of A . Just to be clear, it is important to note the limitations of this notion:

- (1) It is possible that the same element of B is assigned to two (or more) different elements of A . In other words, it is possible that $f(a_1) = f(a_2)$ even though $a_1 \neq a_2$.
- (2) It is possible that there is an element of B which is not assigned to any element of A . In other words, there might exist $b \in B$ such that there does not exist $a \in A$ for which $f(a) = b$.

Example 4. Consider the rule that assigns the element x^2 to every element $x \in \mathbb{R}$. This is an example of a function from $\mathbb{R}$ to itself.

Example 5. Consider the rule that assigns an element $y \in \mathbb{R}$ to an element $x \in \mathbb{R}$ if $x = y^2$. This is not a function because it fails to satisfy the required properties in two ways:

- (1) The element 1 is associated to two distinct elements 1 and -1 . This actually happens for any non-zero positive number. Indeed, every non-zero positive real number has two distinct square roots.
- (2) If x is a negative number in $\mathbb{R}$, there is no y such that $x = y^2$.

Given functions $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, we can define the *composition* of these two functions to be the function $g \circ f : X \rightarrow Z$ which is defined by the formula $g \circ f(x) = g(f(x))$ for every $x \in X$. Note that we cannot compose any two arbitrary functions in general. The composition $g \circ f$ only makes sense if the codomain of f is equal to the domain of g (or is at least contained in the domain of g).

5. INJECTIVE, SURJECTIVE AND BIJECTIVE FUNCTIONS

Let $f : A \rightarrow B$ be a function.

- (1) f is said to be an *injection* (or an *injective function*, or a 1-1 function) if $f(x_1) = f(x_2)$ implies $x_1 = x_2$ for $x_1, x_2 \in A$. (Thus, if $x_1 \neq x_2$, we must also have $f(x_1) \neq f(x_2)$).
- (2) f is said to be a *surjection* (or an *surjective function*, or an *onto function*) if for any $y \in B$, there exists an element $x \in A$ such that $f(x) = y$.
- (3) f is said to be a *bijection* (or an *bijective function*, or a *one-to-one correspondence*) if it is both an injection as well as a surjection.

Examples 6.

- (1) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$ is not an injection because $f(1) = f(-1) = 1$. It is also not a surjection because there does not exist any $x \in \mathbb{R}$ such that $f(x) = -1$.
- (2) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x + 1$ is a bijection.
- (3) Let $\mathbb{Z}$ denote the set of integers. The function $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x) = 2x$ is an injection, but not a surjection.
- (4) Let $f : \mathbb{R} \rightarrow \mathbb{Z}$ be the function $f(x) = \lfloor x \rfloor$ where $\lfloor x \rfloor$ denotes the largest integer less than or equal to x . (This is sometimes called the *floor function*.) This is a surjection, but not an injection.

If $f : A \rightarrow B$ is a *bijection*, we can define a function $g : B \rightarrow A$ by setting $g(y) = x$ if and only if $f(x) = y$. It is easy to verify that g is indeed a function. (You should check that if f is not a bijection, then this definition does not actually give you a

function.) It is called the *inverse function* of f . If g is the inverse function of f , then $g \circ f(x) = x$ for all $x \in X$ and $f \circ g(y) = y$ for all $y \in B$.

Thus, we see that a bijection from a set A to a set B essentially sets up a matching between the elements of A and B . Each element of A is paired up with a unique element of B and each element of B is paired up with a unique element of A .

6. CARDINALITY

We say that two sets *have the same cardinality* if there exists a bijection between them. By the above discussion, it should be clear that if $f : A \rightarrow B$ is a bijection and if A has a finite number of elements, then B should have the same number of elements as A . (This is because f and its inverse function give us a way to pair up the elements of A and B .)

Roughly speaking, “cardinality” could be understood as another word for “number of elements”. For example, the cardinality of the set $\{a, b\}$ is 2.

However, if a set is infinite, its cardinality is not a natural number. It is tempting to think that we could just say that the cardinality is “infinite”. However, it is possible to construct infinite sets which do not have any bijection between them. So they have different cardinalities. This problem can be resolved by generalizing by using the notion of *cardinals*. (See [1, Section 24].) However, we do not need these details for this course.

We will say that a set is *countable* if it admits a bijection with a subset of $\mathbb{N}$. If a countable set does not admit a bijection with $\mathbb{N}$ itself, then we say that it is *finite*. A set that admits a bijection with $\mathbb{N}$ is said to be *countably infinite*. A set that is not countable is said to be *uncountable*.

Given two sets, we should also be able to say which one is “bigger”. In other words, we should be able the cardinalities of the two set (though we have not really defined cardinality here). It seems reasonable to think that if there exists an *injective function* $f : A \rightarrow B$, then the cardinality of A should be less than or equal to that of B . In fact, if f is injective, but not surjective, then one might be tempted to assume that the cardinality of A is strictly “less than” that of B . This is, in fact, true and one can make it all rigorous. We will not really bother do that in this course. However, the following theorem is worth mentioning.

Theorem 7 (Schröder-Bernstein). *Let A and B be sets. Suppose there exist injective functions $f : A \rightarrow B$ and $g : B \rightarrow A$. Then, there exists a bijective function $h : A \rightarrow B$.*

7. READING SUGGESTIONS

There a lot of unproved statements in this lecture. If you are interested, you may read the details in [1], but this is not required for this course.

This lecture roughly corresponds to some portions of Sections 1.2 and 1.5 of the text ([?]).

REFERENCES

- [1] Paul R. Halmos, *Naive Set Theory*, Undergraduate Texts in Mathematics, Springer, 1974. [2](#), [4](#)
- [2] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015. [4](#)